



**Department of Electrical,
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Assignment/Lab Title:	Robot Guidance Challenge
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Introduction

The purpose of this project was to program an HCS12 *eebot* with an MC9S12C32 microprocessor, utilizing Absolute Assembly language to navigate through a maze by following a black line path and to avoid dead ends to move from the start of the maze until the end of the maze.

Theory

The early phase of this project focused on ensuring the photosensors were calibrated accurately to enable the *eebot* to navigate with precision. The *eebot* is comprised of 6 CdS Photoresistor Sensors which are located under the bow of the robot. Coupled with High-Intensity red LEDs, the sensors would work together to illuminate the floor and detect high resistance in darkness and low resistance under light. In doing so, a voltage proportional to the resistance would be generated which would be displayed on an LCD screen located on the top of the robot. These threshold values were essential to allow for accurate readings of both high and low light intensity. Furthermore, The LEDs and sensors are arranged in such a way that they can be utilized to ensure the robot is facing the correct direction at all times. Taking a look at “Figure 2”, it can be seen that there are 3 pairs of both sensors and LEDs, [A-C] to ensure the bot stays on the line going forward, [B-E] and [D-F] to detect if the bot has gone off the line. For example, If the photoresistors noticed a high intensity of light, it would indicate that the robot has gone off course and would tell the main program to call set subroutines to turn accordingly.

Process

In order to complete this project successfully, we utilized several segments of code from all the past labs until now, namely the Battery display from Lab 3 for showcasing our Battery Voltage and even the Current state of the *eebot*, The timer subroutines from Lab 4 to set up the time for each turn, and lastly the entire Dispatcher based FSM along with the motor movement is of a similar structure to the code from Lab 5.

Our *eebot* works on a “Turn-Left” Algorithm where anytime it detects an intersection to it's left, the bot turns, otherwise it continues to move forward. If the bot ever bumps into a wall on its route, it reverses very slightly before making a 180° turn and continues forward as usual. With this method, the bot can reach the end easily despite a few wall bumps that are required to steer it in the correct direction. Lastly, once the bot reaches the end, we can then stop it by simply pushing the back bumper.

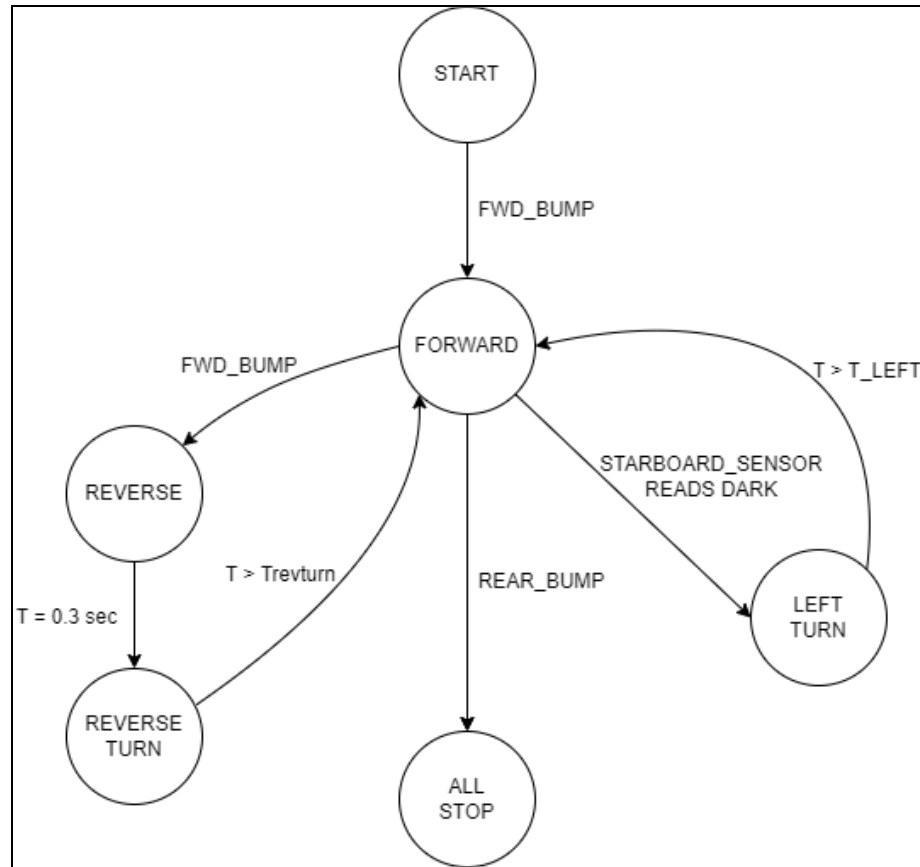


Figure 1, eebot FSM

Challenges

The HCS12 microcontroller was a complicated chip that needed a tremendous amount of code in order for it to function; throughout the project, we encountered a variety of challenges. An example of a challenge was calibrating the photosensors to determine the variance and baseline via trial and error. Throughout the project, the *eebot* was changed which led to a recalibration of the guiding sensors due to each bot having different values. Another challenge was coding the ALL_STOP state, not due to its difficulty but due to a simple error of recognizing the wrong pin.

Appendix

Some important figures provided in the project manual used for the Guidance Challenge:

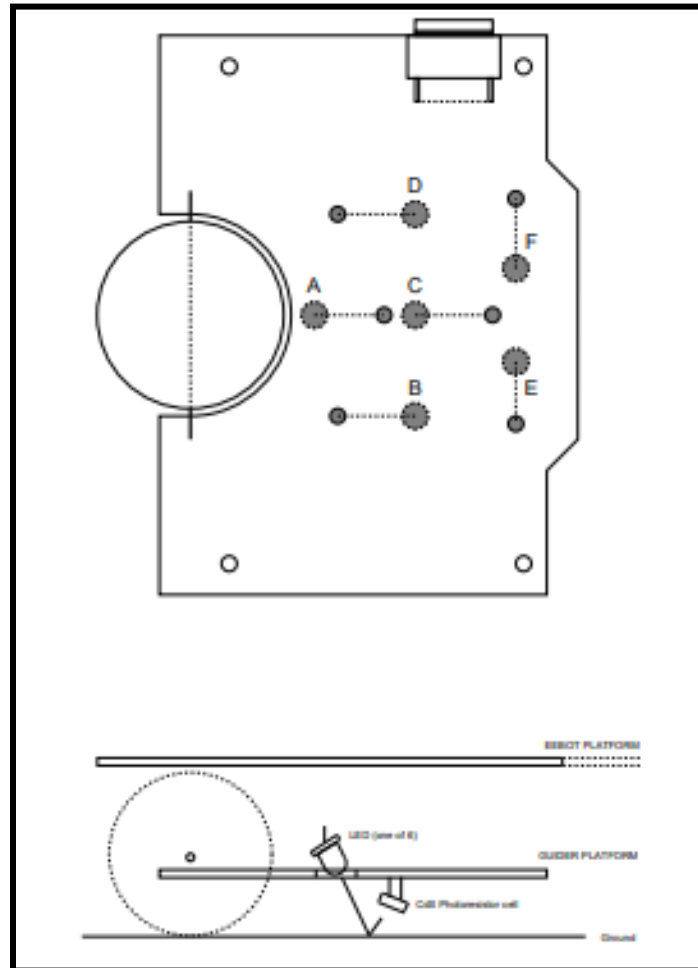


Figure 2, Sensor Locations Top view

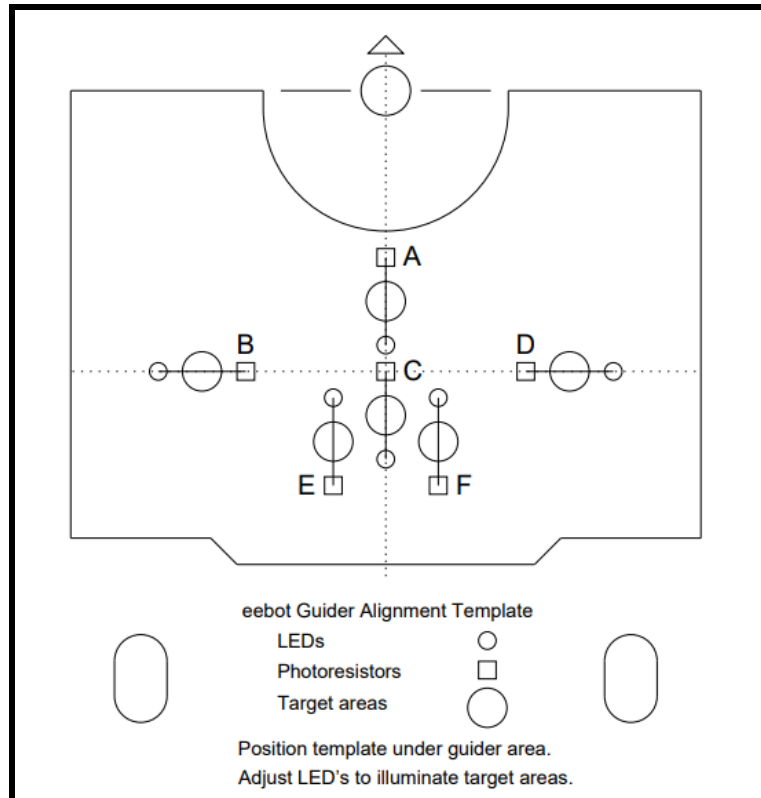


Figure 3, Sensor Template

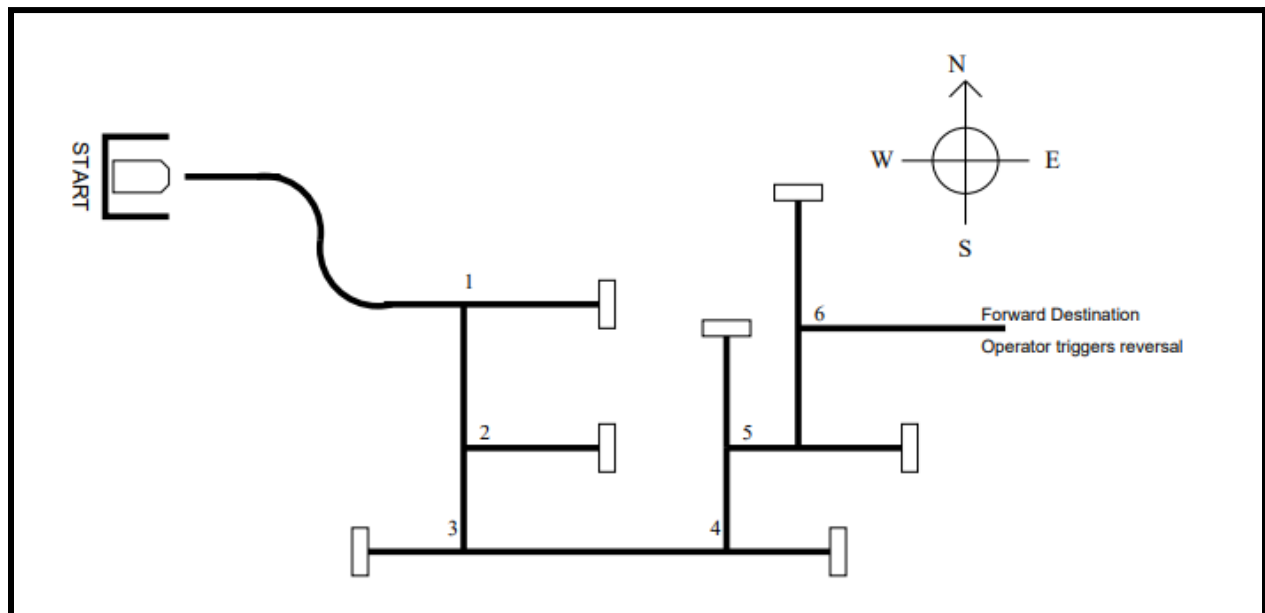


Figure 4, Maze Challenge